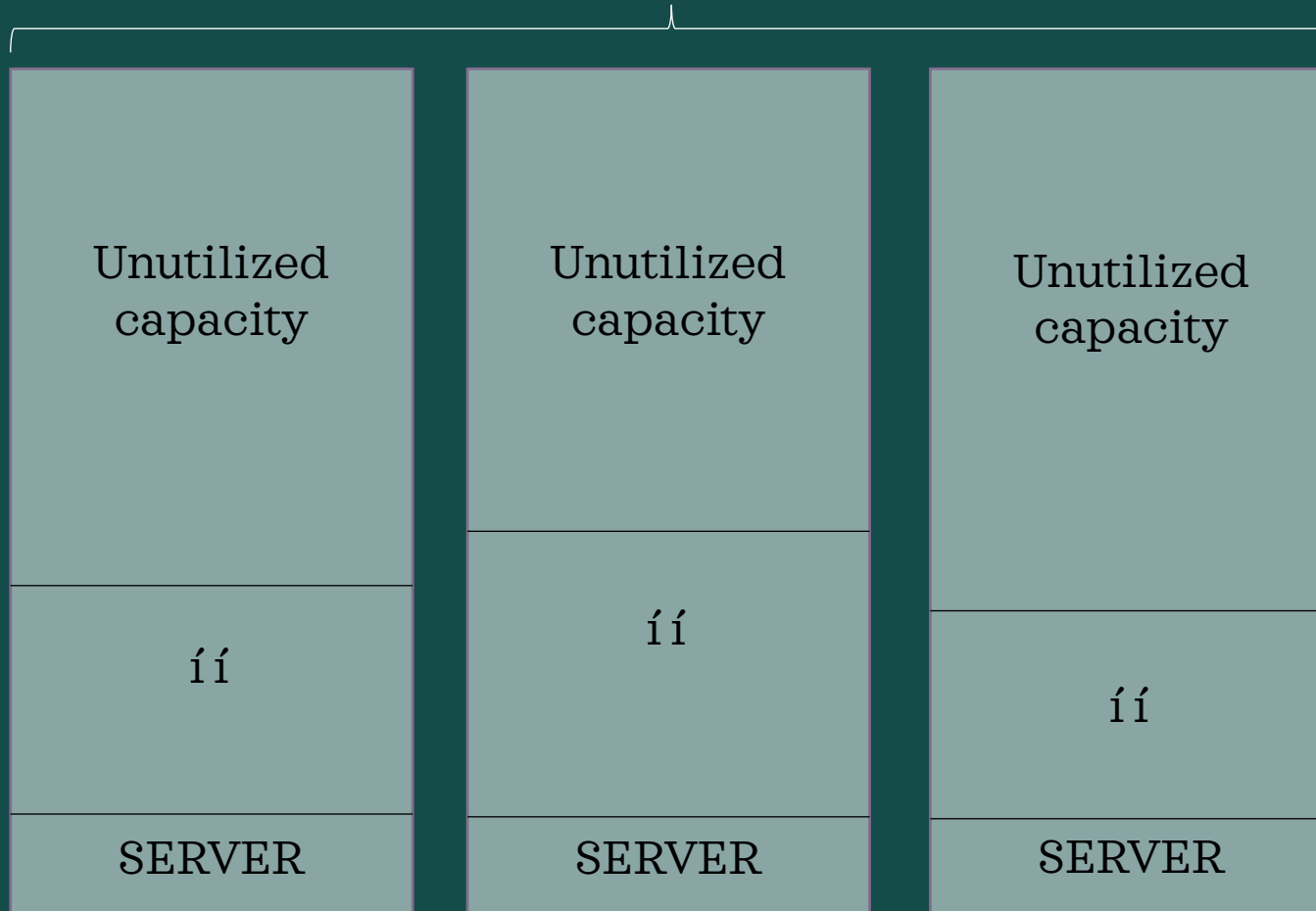


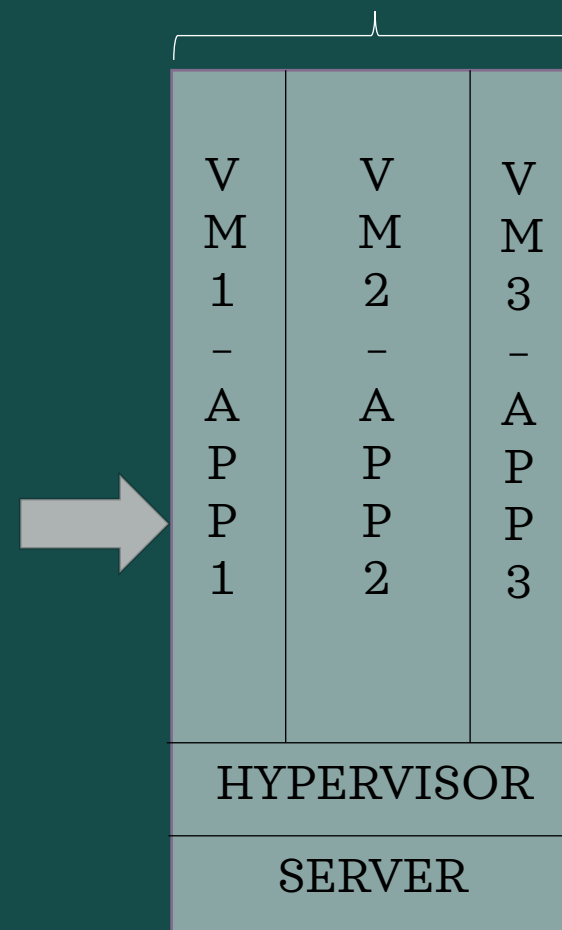
What is a container and how is it different from a VM?

-traditional approach and transition to VMs-

Traditional approach



VMs



Traditional approach:

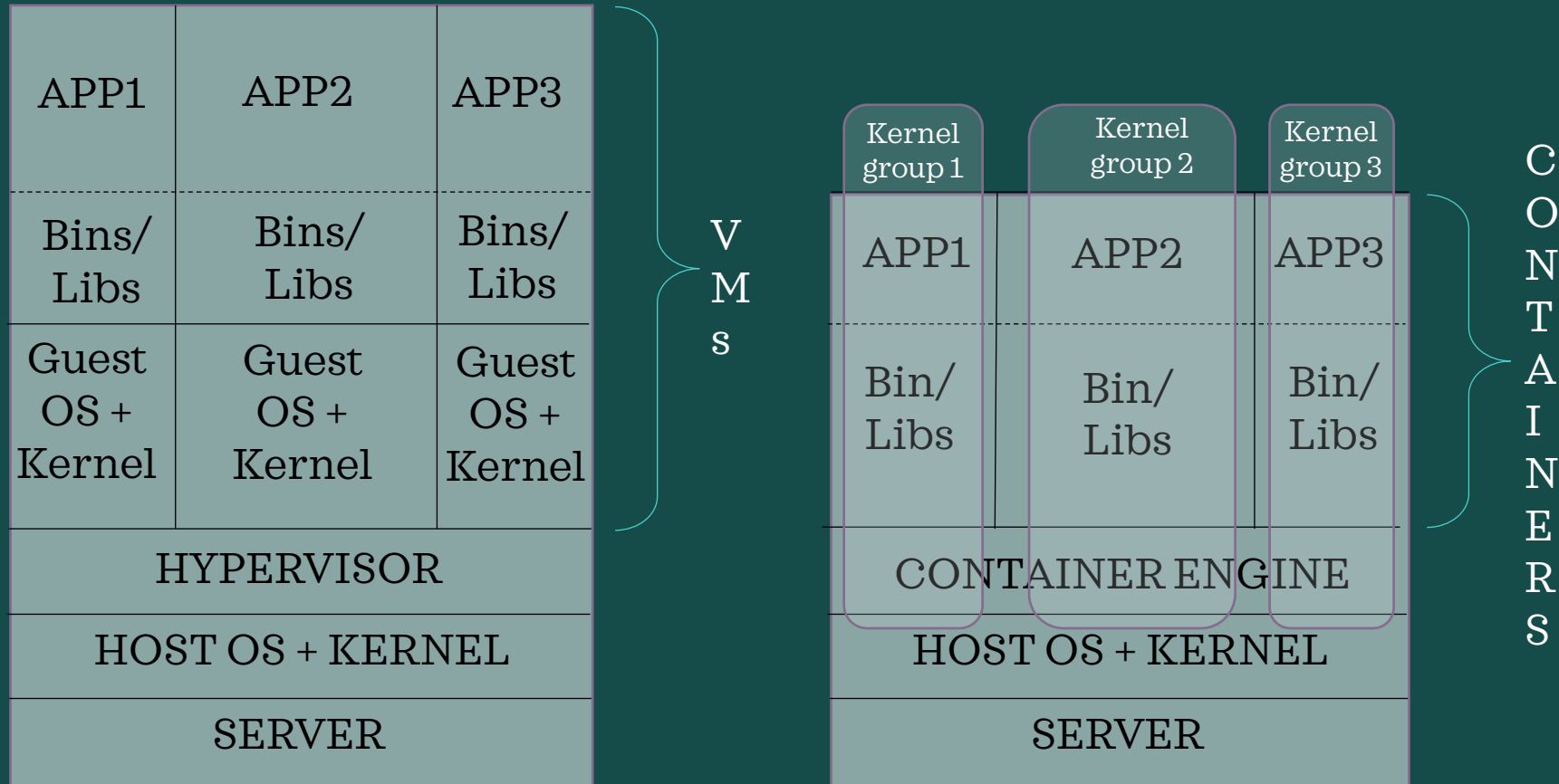
- Each app uses one server resulting in:
 - low usage for servers
 - high cost for hardware

VM approach:

- Software implementation, abstracting the server into multiple VMs resulting in:
 - more efficient usage of servers
 - lower cost

What is a container and how is it different from a VM?

-VMs vs containers-



VM issues:

- Disruptions can happen when multiple apps run on the same VM
- High CPU, RAM, disk consumption from server
- Start-up is slow
- Require licenses

Containers:

- Packages containing everything an app needs to run
- Lightweight, standalone, portable
- Wrapped in Kernel groups and run on the Kernel and OS of the host

What is a container and how is it different from a VM?

-container isolation-

Isolation: accomplished via namespaces-based approach and Control groups (cgroups).

Namespace: layer of isolation by creating separate environments for each container.

Namespace types:

- PID: isolates the process IDs
- Network: isolates the network stack
- Mount: isolates the file system mount points
- User: isolates the user and group IDs
- UTS: isolates the hostname and domain name

Cgroups: restrict resource usage and monitor how those resources are being utilized

What is a container and how is it different from a VM?

-comparison and summary-

Feature/ property	VMs	Containers
Isolation	Full isolation from host OS and other VMs	Lightweight isolate from the host and other containers
Multiple apps	It may lead to conflicts, compatibility issues and disruptions for the apps	Each app packed with its dependencies and isolated from each other
Emulation	Accomplished by hypervisor	Accomplished by container engine
Cost	Usually higher than containers	Usually lower than VMs
Size	~GBs	~MBs
Start-time	~Minutes	~Seconds
Portability/ replaceability	Difficult	Easy
Fit for monolithic/ microservices	Support monolithic apps	Good fit for microservices architecture